

## NVIDIA open sources KAI Scheduler to encourage community collaboration for innovation in AI infrastructure

NVIDIA has open sourced the KAI Scheduler, a Kubernetes-native GPU scheduling solution originally developed within the Run:ai platform. This move aims to foster community collaboration and innovation in AI infrastructure.

KAI Scheduler continuously recalculates fair-share values and adjusts quotas and limits in real-time, automatically matching current workload demands.

By combining gang scheduling, GPU sharing, and a hierarchical queuing system, the scheduler reduces wait times for compute access, allowing tasks to launch promptly as resources become available.

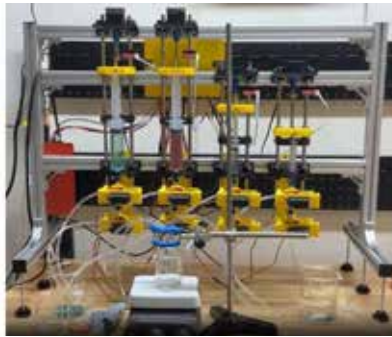
It enforces resource guarantees, ensuring that AI practitioner teams receive their allocated GPUs while dynamically reallocating idle resources to other workloads, thus preventing resource hogging and promoting overall cluster efficiency.

The scheduler features a built-in podgroup that automatically detects and connects with AI tools and frameworks like Kubeflow, Ray, Argo, and the Training Operator, reducing configuration complexity and accelerating development.

NVIDIA is encouraging enterprises, startups, research labs, and open source communities to experiment with the scheduler and contribute their learnings. The KAI Scheduler is available under the Apache 2.0 licence, and the community is invited to join the project on GitHub to collaborate and help shape the future of AI workload orchestration.

Device), a groundbreaking open source robotic system designed for automating chemical experiments. Built using a 3D printer and affordable off-the-shelf electronic components, FLUID aims to make laboratory automation more accessible and customisable.

The team demonstrated the system's capabilities by automating the co-precipitation of cobalt and nickel, an important process in materials science. This was executed with remarkable precision and efficiency. "By using open source tools, 3D printing, and widely available electronics, we created a cost-effective



and adaptable robotic system tailored for specific lab needs," said Mikael Kuwahara, the study's lead author.

FLUID has four independent modules, each with a syringe, two valves, a servo motor for valve control, and a stepper motor to control the syringe plunger. Each module also has an end-stop sensor to detect the syringe's maximum fill position.

These modules are connected to microcontroller boards that receive commands from a computer via USB. The system also includes software that lets users control the robot's functions, such as valve adjustments and syringe movements.

The researchers plan to integrate additional sensors to monitor other parameters, such as temperature and pH. These will help the robot to handle a wider variety of chemical reactions, including polymer mixing and organic synthesis. The software will also be enhanced to include features like macro recording to streamline repetitive tasks.

## New version of Calico Open Source boosts Kubernetes security

Tigera has unveiled Calico Open Source 3.30, marking its most significant functionality upgrade to date. This latest release enhances observability, strengthens security, and expands support for managing ingress traffic in Kubernetes environments. With version 3.30, the company is bringing enterprise-grade security and observability tools from its commercial editions to open source users.

It introduces several key enhancements that significantly improve observability, security, and ingress management in Kubernetes environments. One of the standout features is advanced observability with Goldmane, a gRPC-based API that streamlines access to flow logs and metrics. This provides DevOps teams with deeper visibility into service-to-service communications and workload-specific data, reducing troubleshooting time from days to minutes. Complementing this is Whisker, a web-based UI integrated with Goldmane, which allows users to filter flow logs, analyse metadata, and resolve connectivity issues more efficiently. These tools not only enhance network troubleshooting but also improve security monitoring and policy management by offering real-time insights into traffic behaviour.

Security policies are essential for safeguarding Kubernetes workloads but enforcing them without prior testing can lead to disruptions. To address this, the new version introduces GlobalStagedNetworkPolicy and StagedNetworkPolicy, enabling developers to test and audit security policies before they are activated. These staged policies appear in flow logs and simulate real-world conditions, allowing organisations to validate their policies without impacting live environments.

Ingress management also sees a major upgrade with the introduction of the Calico Ingress Gateway, a robust, vendor-neutral implementation based on Envoy Gateway. This solution provides improved load balancing, failover strategies,